

Ergo 3: Virtual Globetrotter

Design Document

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1. Introduction

Project Vision

The *Virtual Globetrotter* project aims to provide children with an immersive and interactive experience, allowing them to explore different locations around the world using an interactive map and AI-generated images. The project utilizes speech recognition technology to create a unique and engaging experience, while also promoting essential media literacy skills. The app is designed with a simple and intuitive interface, focusing on smooth navigation, helpful feedback, and colourful visuals, making it accessible to anyone.

Target Audience

The main audience for this project is children, particularly those who may not have the opportunity to explore the world due to various restrictions. The app is designed with accessibility features, including:

- Speech recognition, allowing users to interact with the app using voice commands
- Large, easy-to-click buttons
- Simple and intuitive navigation
- Clear, real-time feedback for every action

2. Look and Feel

Colours

The app features a vibrant and engaging colour scheme, with the following main colours:

-  **Primary Gradient:** Linear gradient from Bright Blue (#3b82f6) to Purple (#9333ea)
-  **Button Hover Gradient:** Linear gradient from Darker Blue (#2563eb) to Deep Purple (#7c3aed)
-  **Glass Panels:** Semi-transparent White (rgba(255, 255, 255, 0.1)) with blurred backgrounds
- **Text Colors:**
 -  Titles: White (#FFFFFF)
 -  Subtitles: Light White (rgba(255, 255, 255, 0.9))
 -  Interactive Elements: Light Aqua (#d2e4ef)
-  **Progress Bar:** Green (#4CAF50)

Each continent on the map is also assigned a unique colour to enhance clarity:

-  Asia: Light Red (#FF9999)
-  North America: Soft Aqua (#66CDAA)
-  Africa: Light Green (#90EE90)
-  South America: Warm Orange (#FFE5B4)
-  Europe: Sky Blue (#87CEEB)
-  Australia: Sandy Brown (#DEB887)

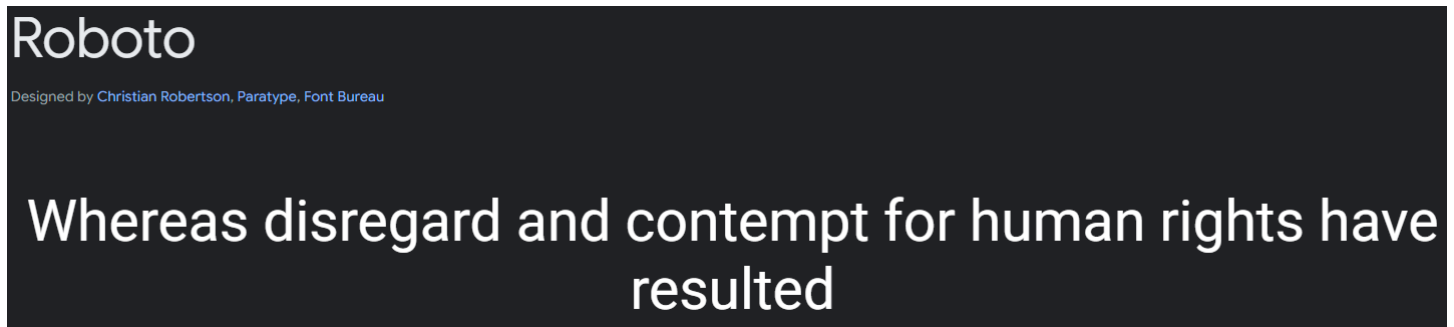
The colour scheme is designed to be visually appealing, easy to read, and consistent throughout the app.

Text

The app uses the clean and simple **Roboto** font, with the following sizes:

- Titles: Scalable (e.g., 4rem on large screens).
- Subtitles: 1.5rem, with scalable adjustments.
- Buttons: 1.25rem.
- **Special Effects:** Subtle fade-in animations for headings and subtitles

The text is designed to be clear, concise, and easy to read, with sufficient contrast between the text and background.

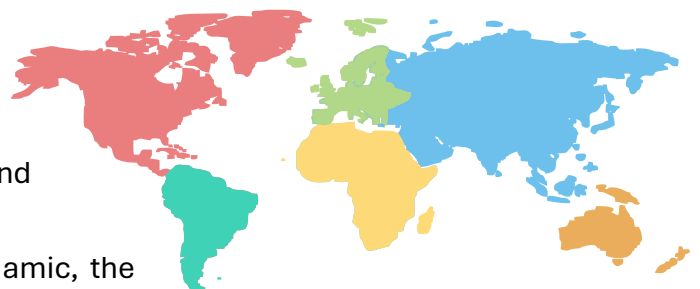


Source: [Roboto - Google Fonts](#)

Graphics

The world map is created using an SVG file, featuring:

- Easy-to-click areas for each continent
- White borders to separate regions
- Interactive elements, such as hover effects and animations, to enhance the user experience



To make the UI even more visually appealing and dynamic, the following custom animations were added:

- Particles: Floating white dots (8px) for background ambiance.
- Buttons: Bounce animations with smooth hover transitions.

A lot of different images are used as a background for the web application. We will use self-generated images for all the slideshows and the background, using our AI pipeline code. This ensures that the application will not get any copyright claims if deployed.

Lastly, Howest MCT and Ergotherapy courses logo were added to show that the project was made in collaboration between the two departments.



3. Functionality

Layout

The app's layout is designed to be intuitive and easy to navigate, with:

- Each screen filling the whole window, with a minimum width of 1024px and a minimum height of 768px
- Centred elements for balance, with a maximum margin of 20px
- Compatibility with both large and small screens, including desktop, tablet, and mobile devices

The layout is designed to be flexible and adaptable, with a responsive design that adjusts to different screen sizes and orientations.

Important Features

The app features:

- Big and colourful buttons with rounded edges, that are scalable based on the user's device
- Loading screen with a progress bar with smooth transitions and slideshow of AI-generated images
- Voice input with real-time display of spoken words, with a speech recognition system that supports English language
- Server communication that sends user choices to a local server and handles any issues smoothly, with error handling and feedback mechanisms in place

User Feedback

The app provides real-time feedback, including:

- Text display of spoken words, with a maximum delay of 1 second
- Voice recording button changes when listening, with a visual indicator that shows when the app is listening
- Smooth transitions between buttons and pages, with a maximum transition time of 2 seconds

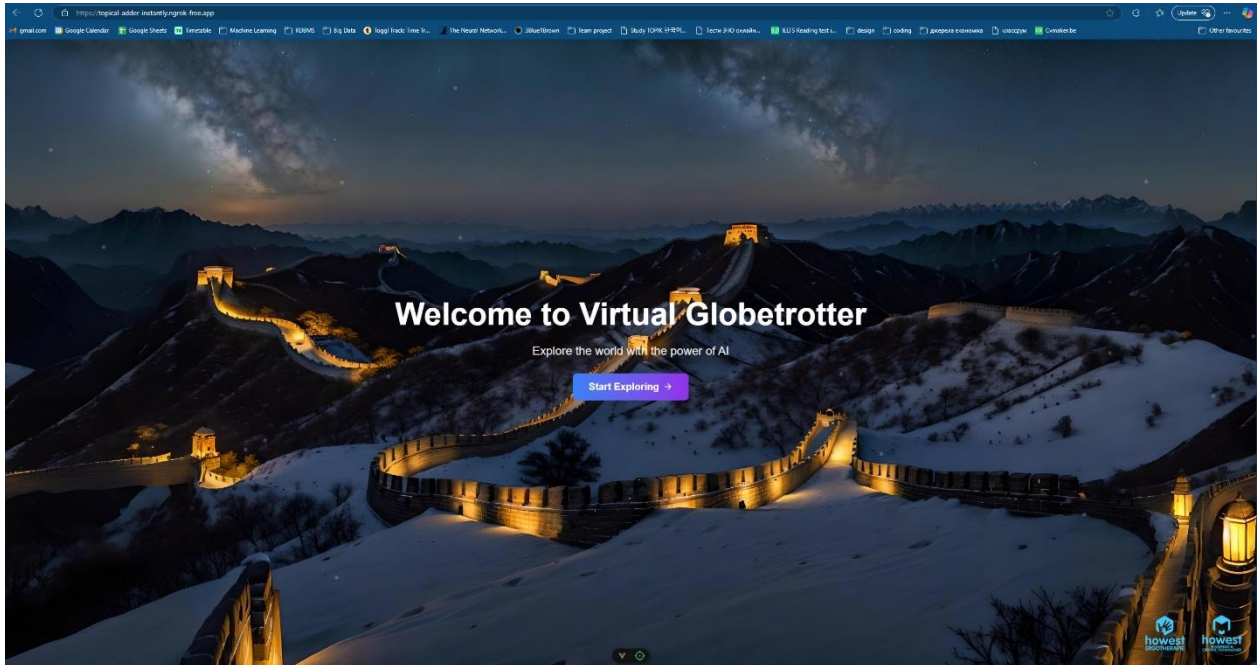
The app also provides feedback mechanisms, such as error messages and alerts, to inform the user of any issues or errors that may occur.

4. Project Details

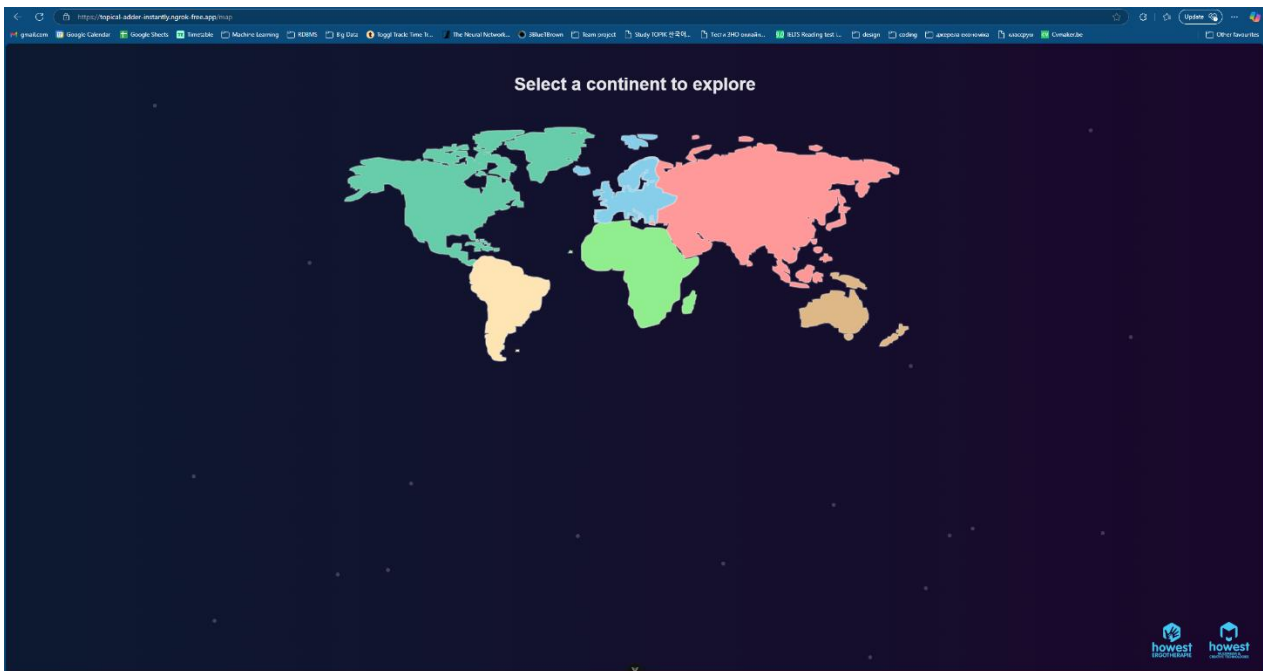
Views

The app features the following views:

- **Home:** start your journey, with a welcome message and a call-to-action to start exploring



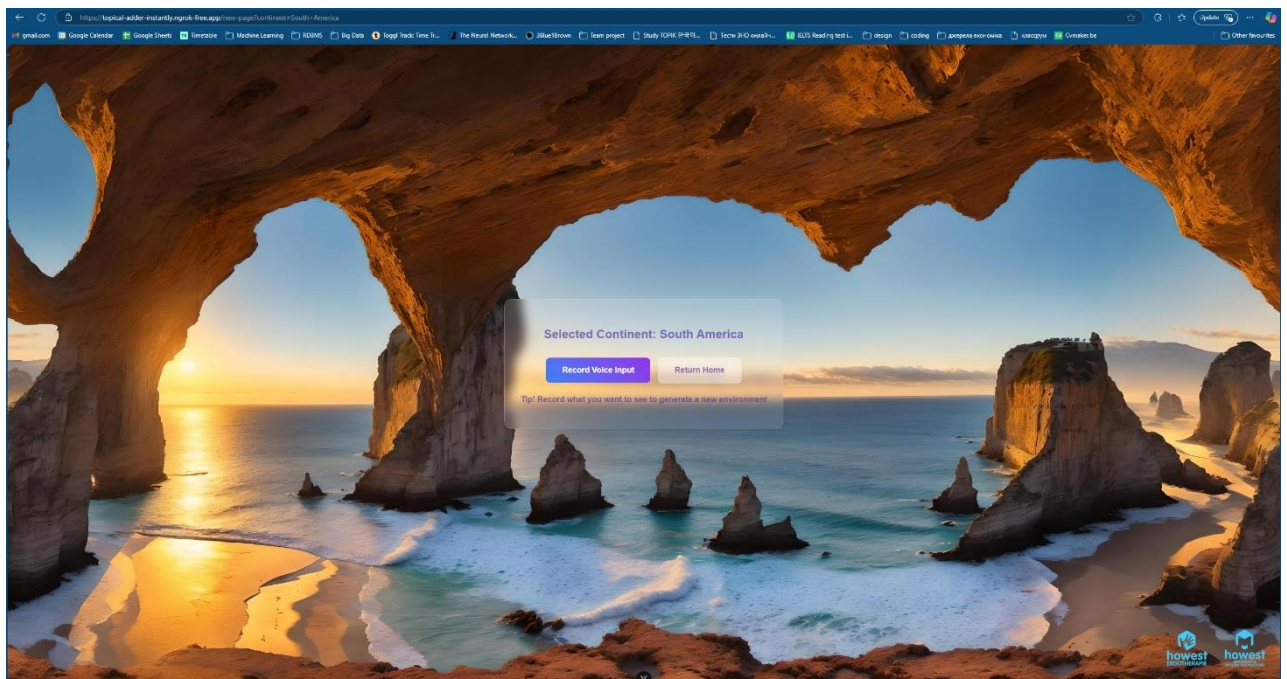
- **Map:** choose a continent, with an interactive map that allows users to select a continent and view its corresponding images



- **Loading:** see progress while waiting, with a loading animation, a progress bar, and a slideshow in the background.



- **New Page:** speak and see changes, with a voice command interface that allows users to interact with the app using speech



Tools and sources

The app is built using accessible and beginner-friendly tools:

- **Vue.js:** A flexible JavaScript framework used for building the app with reusable components. It allows for a clear structure and styling, ideal for new developers
- **Vue Router:** A library for navigating between pages without reloading the app. This makes transitions smooth and enhances user experience
- **Axios:** A simple tool for communicating with the server. It sends and receives data, making features like loading new maps or handling voice inputs possible
- **Web Speech API:** A built-in browser feature for recognizing and processing voice input. This lets users interact hands-free and supports various accents
- **Custom SVG Maps:** A scalable and clickable world map, designed to engage users. Each continent is interactive, making exploration fun and educational. [Original map](#) was taken and manually changed to the SVG file, which was then used to create a world map page

5. Inspiration

The idea for "Virtual Globetrotter" came from tools and designs that make learning and exploring simple and engaging. Real-world examples include:

- **United Nations' Sustainable Development Goals:** Goals such as education and accessibility shaped the focus on inclusivity and user-friendly design.
- **AI-Generated visuals:** By browsing user results on a platform called [Civitai](#) which calls itself "The Home of Open-Source Generative AI", we gained some insights about how we should construct our prompt and what results can we expect.

Inspiration also came from:

- **Educational Apps:** Games like "Kahoot!" and "Duolingo" showed how to make learning fun and interactive.
- **Accessibility Tools:** Features like large buttons and voice control were designed with real-world tools like screen readers and voice assistants in mind.

Links:

GitHub

<https://github.com/IsabelleDesle/VirtualGlobeTrotter>

Trello

[ERGO3: Virtual Globetrotter | Trello](#)